Session 4 Labs: Accessible forms and your project in HTML

CM-501 Frontend Web Development · EPA 2

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Warm-up · What will it say? (teams of 2–3, 10 min)

For each snippet write what a screen reader announces: **name, role** (e.g. “Send, button”). If there is no name, write “no name”.

	Snippet	Announced	Problem?
A	`<button>Send</button>`		
B	`<button><svg viewBox="0 0 24 24">…</svg></button>`		
C	`<button aria-label="Close"><svg aria-hidden="true">…</svg></button>`		
D	`<img src="special.png">`		
E	`<input type="email" placeholder="E-mail">`		
F	`<label for="n">Name</label> <input id="n" aria-label="Your full name">`		
G	`<div onclick="send()">Send</div>`		
H	`<nav aria-label="Main navigation"> … </nav>`		

Lab F · Form surgery (individual, neighbours may help, 22 min)

Files: clone once `git clone https://github.com/sr-mudt/cm501-labs.git`, then `git pull` before every session. Today: `cm501-labs/session-04/workshop-booking-broken.html`: the booking form for the Latte Art Workshop.

1. Test first (5 min). Keyboard only (`Tab`, `Shift+Tab`, `Space`, arrow keys, `Enter`) and with the screen reader. What fails?

Keyboard: Tab never reaches BOOK (a <div onclick>, not focusable, Enter/Space do nothing) -> cannot book without a mouse. Focus is invisible (*:focus { outline: none }). Both contact radios can be ticked and arrows don't move between them (different names: contact / contact2). "?" hint is a mouse-only title tooltip.

Screen reader: fields only read as "edit text" (placeholder, no label); select reads just "1, popup button"; radios/checkboxes have no question or option names; link reads "here"; required is only a red star; "Invalid input!" shown before anything was typed. No lang, title "Form", no h1, no main.

2. Repair it. Checklist (in order of importance; if time runs out, stop after the button and the focus):

- [x] `<html lang="en">`, meaningful `<title>`, viewport meta tag, a real `<h1>`, content inside `<main>`
- [x] every field has a **visible** `<label>` connected with `for` / `id` (placeholders are not labels)
- [x] the right `type`: `email`, `tel` … and `autocomplete` for name, e-mail, phone (never `off`)

- [x] **required** fields: `required` attribute **and** “(required)” in the label text (not only a red star)
- [x] the e-mail **hint** is visible text, connected with `aria-describedby` (the `title` tooltip goes)
- [x] the **select** has a label
- [x] radio buttons: **one** `name` for the group, `<fieldset>` + `<legend>` with the question, a `label` per option
- [x] checkboxes: `<fieldset>` + `<legend>`, a `label` per option
- [x] privacy checkbox: label, link text that makes sense without “here”
- [x] the generic “Invalid input!” goes (errors come in Lab F2)
- [x] a real `<button type="submit">` with a clear text, e.g. “Book my place”
- [x] **visible focus** (remove `outline: none`, add a `:focus-visible` style)

3. **Test again.** Same tests. What does the screen reader say now on …

Field	Announced
E-mail	"E-mail (required), required, edit text, We send the confirmation to this address."
“How should we contact you?” (first radio button)	"E-mail, radio button, 1 of 2, How should we contact you? (required), group" - arrow keys now move within the group
Book button	"Book my place, button" - reachable with Tab, Enter submits

Lab F2 · Build the error state (15 min)

Copy your repaired form to `workshop-booking-error.html`. Build the page **as it looks after a failed submit**: the e-mail field contains `anna.beispiel`, the privacy box is not ticked.

- ☐ an **error summary** at the top: heading "There are 2 problems with your booking", a list with **links** to both fields, `role="alert"`, `tabindex="-1"`
- ☐ next to each faulty field a **message** that says what is wrong **and** how to fix it, starting with "Error:"
- ☐ the message has an `id`; the field has `aria-describedby="…"` pointing to it (and to the hint, if there is one)
- ☐ the field has `aria-invalid="true"`
- ☐ the typed value is **kept** (`value="anna.beispiel"`)
- ☐ the error is visible **without colour** (text, icon, thicker border)

Test: what does the screen reader announce on the e-mail field?

Lab G · Your project's HTML baseline (individual, 8 + 22 + 7 min)

1 · My project (3 min alone, 3 min in pairs)

Pitch: *A website for … so that … can …*

Audience (who, on which device, what task?)

4 pages (name + one line each)

The form (what does it ask for?)

What changes (dates, events, a list …)

Real or fictional? (real: permission asked?)

Partner asks: "Who uses it, and for what?" Note one thing to sharpen: _____

2 · Build (22 min)

1. Copy `cm501-labs/session-04/project-starter/` into the **root** of your project repository · commit *"Add HTML baseline starter"*
2. `index.html`: replace every TODO (site name, `h1`, real text, navigation for your 4 pages)
3. one more page from `page-template.html` with real content; the other pages as **stubs** (heading + one sentence)
4. fill in the README table: **where** is each feature? · commit and push

Example of a finished baseline: `cm501-labs/session-04/project-example/` (Isartal Climbing Club, fictional).

The checklist (also in the README of the starter)

#	Must	✓
M1	`lang`, `charset`, `viewport`, unique `title` (“Page · Site”), `meta description`, favicon	
M2	landmarks `header` `nav` `main` `footer`; skip link as the first element	
M3	navigation as a list, identical on all pages, `aria-current="page"` on the current page	
M4	one `h1` per page, no skipped heading levels	
M5	images: `alt` (or `alt=""`), `width`/`height`, `loading="lazy"` below the fold; one `figure` + `figcaption`	
M6	lists for lists; `article` / `section` with headings where they fit	
M7	link texts that make sense alone; `<address>` with `tel:` / `mailto:` links	
M8	`<time datetime="…">` for at least one date or time	
M9	one form: visible labels, right `type`, `autocomplete`, “(required)” + `required`, real `<button>`; `fieldset`/`legend` for groups	
M10	validator 0 errors · axe 0 violations · keyboard: everything reachable, focus visible	

#	Optional: pick what fits your content	✓
O1	responsive image: `srcset` + `sizes` or `<picture>`	
O2	data table with `<caption>` and `<th scope>`	
O3	`<details>` / `<summary>` (e.g. FAQ)	
O4	`<dl>` for name–value pairs (opening hours, facts)	
O5	`<blockquote>` with source; `<abbr>` with the expansion written out once	
O6	icon-only link or button: `aria-label`, `aria-hidden="true"` on the icon	
O7	`<iframe>` with `title` (map, video); privacy considered	
O8	`lang` on a passage in another language	

3 · Check your baseline: tool + human (7 min)

Test	Result
Validator (validator.w3.org/nu)	errors:
axe DevTools	violations:
Keyboard: skip link first? all reachable? focus visible?	
Screen reader / DevTools Accessibility: headings, landmarks, form fields	
What did only I find, not the tools?	

Due before Session 5 (prerequisite for EPA 3): baseline pushed, topic and table in the README. A classmate checks it at the start of Session 5.

Activity · Does the BFSG apply? (plenary, 5 min)

For each case: **yes** / **no** / **other law**, and one reason.

#	Case	Answer	Reason
1	Online shop of a sneaker start-up, 25 employees		
2	Website of a café with menu and opening hours, 6 employees		
3	Banking app of a large bank		
4	Portfolio site of a freelance web designer		
5	Ticket booking of a regional bus company		
6	Course registration portal of a public university		
7	Wholesale portal only for business customers		
8	E-book shop of a large publisher		

Homework · Repair Webword (individual, ≈ 30 min, no points, before Session 5)

File: `cm501-labs/session-04/webword-broken/` (copy the folder before you change it): our game **after two guesses**, built the way many word games on the web are built. It works with a mouse. Does it work for everybody? **Do not look at the live version** (sr-mudt.github.io/webword) before you are done: you compare at the end.

1. Test first (10 min). Mouse away. Keyboard only, then the screen reader, then look at it as if you could not tell blue from orange. Run axe as well.

Test	What fails?
Keyboard: can you reach "?", GO, the keys?	
Screen reader: what do you hear for the board?	
Without colour: which letters are correct?	
axe: how many issues? Which did it miss?	

2. Repair it (20 min). Native HTML first, ARIA only where HTML is not enough (slide 17).

- ☐ `lang`, viewport, meaningful `title`; `header`, `main`, a real `h1`
- ☐ the "?" becomes a **button** with a name; the help opens in a native `<dialog>` or `popover` (Esc closes it)
- ☐ the board is a **list of guesses**; each guess has a **text summary** for screen readers ("Guess 1, PROPS: P correct, …"), the tiles are `aria-hidden`
- ☐ the state is visible **without colour**: a symbol, a pattern, strike-through; contrast ≥ 4.5 : 1 on every tile
- ☐ the input has a **visible label**; GO becomes a real **submit button** in a `<form>`
- ☐ the message line is a **live region** (`role="status"`) and does not vanish
- ☐ the keyboard keys are **buttons**; ⌫ gets a name; a used key says its state (`aria-label="R, correct"`)
- ☐ visible **focus**; remove the useless `aria-label` on the board `div`
- ☐ the help text explains the marks **without** "blue = …, orange = …"

3. Compare. Open the live version, view its source, and write down **one thing it solved differently** from you:

_____

Your repaired HTML can be the starting point of the Grid bonus in Session 5.